

Body Mass Index (BMI)

Introduction to BMI

- Body Mass Index (BMI) is a numerical value derived from a person's weight and height.
- It is a simple and widely used screening tool to categorize body weight.
- BMI helps identify whether an individual is underweight, normal weight, overweight, or obese.

History and Origin of BMI

- BMI was developed by Belgian mathematician Adolphe Quetelet in the 19th century.
- Originally called the Quetelet Index.
- Later adopted by the World Health Organization (WHO) for public health assessments.

Importance of BMI

- Provides a quick assessment of body weight status.
- Helps in identifying health risks associated with excess or low body weight.
- Used in clinical practice, epidemiological studies, and public health policies.

BMI Formula

- BMI = Weight (kg) / [Height (m)]² - **Metric Units**
- Alternatively: BMI = (Weight in pounds × 703) / [Height in inches]² - **Imperial Units**

Example of BMI Calculation

- Weight = 70 kg
- Height = 1.70 m
- BMI = $70 / (1.70 \times 1.70) = 24.22 \text{ kg/m}^2$

- A Patient's weight is 160 lbs and height is 4 ft 8 inches. Calculate BMI.
- Weight = 160 pounds
- Height = 4 ft 8 inches
- 1 ft = 12
- So height = 48 inches + 8 inches = 56 inches
- $BMI = 160 \times 703 / 3136 = 35.86$

- **Activity : Compute your own BMI value.**

WEIGHT	lbs	90	100	110	120	130	140	150	160	170	180	190	200	210	220	230	240	250	260	270	280	290	
	kgs	41	45	50	54	59	64	68	73	77	82	86	91	95	100	104	109	113	118	122	127	132	
HEIGHT			Underweight					Healthy					Overweight				Obese				Extremely Obese		
	ft/in		cm																				
4'8"	142.2	20	22	25	27	29	31	34	36	38	40	43	45	47	49	52	54	56	58	61	63	65	
4'9"	144.7	19	22	24	26	28	30	32	35	37	39	41	43	45	48	50	52	54	56	58	61	63	
4'10"	147.3	19	21	23	25	27	29	31	33	36	38	40	42	44	46	48	50	52	54	56	59	61	
4'11"	149.8	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48	51	53	55	57	59	
4'12"	152.4	18	20	21	23	25	27	29	31	33	35	37	39	41	43	45	47	49	51	53	55	57	
5'1"	154.9	17	19	21	23	25	26	28	30	32	34	36	38	40	42	43	45	47	49	51	53	55	
5'2"	157.4	16	18	20	22	24	26	27	29	31	33	35	37	38	40	42	44	46	48	49	51	53	
5'3"	160.0	16	18	19	21	23	25	27	28	30	32	34	35	37	39	41	43	44	46	48	50	51	
5'4"	162.5	15	17	19	21	22	24	26	27	29	31	33	34	36	38	39	41	43	45	46	48	50	
5'5"	165.1	15	17	18	20	22	23	25	27	28	30	32	33	35	37	38	40	42	43	45	47	48	
5'6"	167.6	15	16	18	19	21	23	24	26	27	29	31	32	34	36	37	39	40	42	44	45	47	
5'7"	170.1	14	16	17	19	20	22	24	25	27	28	30	31	33	34	36	38	39	41	42	44	45	
5'8"	172.7	14	15	17	18	20	21	23	24	26	27	29	30	32	33	35	37	38	40	41	43	44	
5'9"	175.2	13	15	16	18	19	21	22	24	25	27	28	30	31	33	34	35	37	38	40	41	43	
5'10"	177.8	13	14	16	17	19	20	22	23	24	26	27	29	30	32	33	34	36	37	39	40	42	
5'11"	180.3	13	14	15	17	18	20	21	22	24	25	27	28	29	31	32	33	35	36	38	39	40	
6'0"	182.8	12	14	15	16	18	19	20	22	23	24	26	27	28	30	31	33	34	35	37	38	39	
6'1"	185.4	12	13	15	16	17	18	20	21	22	24	25	26	28	29	30	32	33	34	36	37	38	
6'2"	187.9	12	13	14	15	17	18	19	21	22	23	24	26	27	28	30	31	32	33	35	36	37	
6'3"	190.5	11	13	14	15	16	18	19	20	21	23	24	25	26	28	29	30	31	33	34	35	36	
6'4"	193.0	11	12	13	15	16	17	18	19	21	22	23	24	26	27	28	29	30	32	33	34	35	
6'5"	195.5	11	12	13	14	15	17	18	19	20	21	23	24	25	26	27	28	30	31	32	33	34	
6'6"	198.1	10	12	13	14	15	16	17	18	20	21	22	23	24	25	27	28	29	30	31	32	34	
6'7"	200.6	10	11	12	14	15	16	17	18	19	20	21	23	24	25	26	27	28	29	30	32	33	
6'8"	203.2	10	11	12	13	14	15	16	18	19	20	21	22	23	24	25	26	27	29	30	31	32	
6'9"	205.7	10	11	12	13	14	15	16	17	18	19	20	21	23	24	25	26	27	28	29	30	31	
6'10"	208.2	9	10	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	
6'11"	210.8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	25	26	27	28	29	30	

BMI Classification for Adults (WHO)

- Underweight: BMI < 18.5
- Normal weight: BMI 18.5 – 24.9
- Overweight: BMI 25.0 – 29.9
- Obese: BMI $\geq$ 30.0

Obesity Classification

- Class I Obesity: BMI 30.0 – 34.9
- Class II Obesity: BMI 35.0 – 39.9
- Class III Obesity (Severe): BMI ≥ 40.0

BMI for Children and Adolescents

- BMI calculation is similar to adults.
- Interpretation is based on age- and sex-specific percentile charts.
- Categories include underweight, healthy weight, overweight, and obese.

Health Risks of Low BMI

- Malnutrition and nutrient deficiencies.
- Weakened immune system.
- Increased risk of osteoporosis and anemia.

Health Risks of High BMI

- Increased risk of cardiovascular diseases.
- Type 2 diabetes (**Type 2 diabetes** happens when the body cannot use insulin correctly and sugar builds up in the blood) and metabolic syndrome.
- Joint problems, sleep apnea, and certain cancers.

Limitations of BMI

- Does not distinguish between fat mass and muscle mass.
- May misclassify athletes or muscular individuals.
- Does not account for age, sex, or fat distribution.

BMI vs Other Measures

- Waist circumference assesses abdominal fat.
- Waist-to-hip ratio indicates fat distribution.
- Body fat percentage provides a more accurate body composition.

BMI in Public Health

- Used to monitor population-level obesity trends.
- Helps design health policies and intervention programs.
- Aids in identifying at-risk populations.

- BMI is a simple, cost-effective screening tool.
- It provides useful insight into weight-related health risks.
- Should be interpreted alongside other clinical assessments.